The Roles of Policy, Labor, and Perception of Opioid Misuse Risk on Opioid Demand

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Motivation

- Opioid misuse: using opioids not as directed by a doctor.
  - People misuse opioids for analgesic effects and rewarding properties.
  - Opioid misuse increases the risk of Opioid Use Disorder and overdose deaths.
  - Using illegally obtained opioids is riskier than misusing prescribed opioids.
- Laws and policies on prescribing opioids have been imposed recently.
  - The purpose is to decrease opioid misuse and, thus, opioid overdose deaths.
  - Reason #1: Opioids are too commonly prescribed, facilitating misuse.
  - Reason #2: People are over-confident about opioid misuse risk.

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Motivation

- These policies increased overdose deaths, but decreased opioid misuse.
- Overdose deaths were increasing due to illegal opioids.
- Opioid misuse was decreasing in all types of misuse.

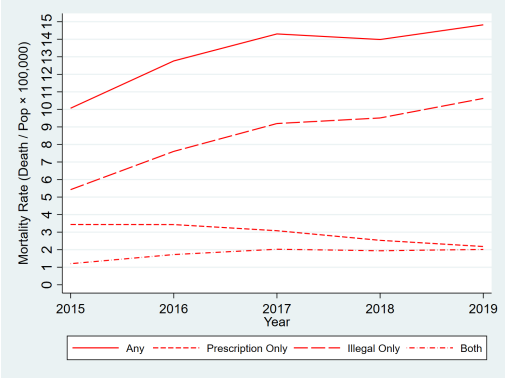

Figure: Opioid Mortality Rates by Type

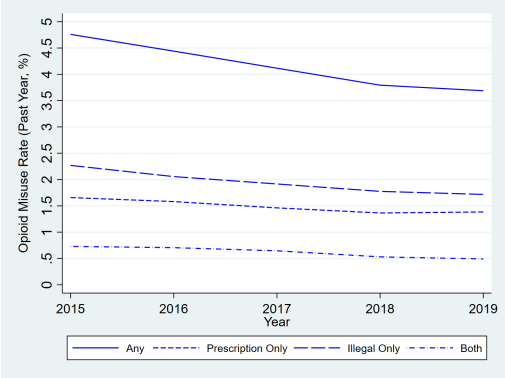

Figure: Opioid Misuse by Rates Type

Research Question

- Who are the winners and losers of state-level policies on prescribing opioids?
  - How strong is the substitution effect toward illegal opioids?
  - Is there a heterogeneity in substitution patterns across socioeconomic factors?
  - How much does the perception of opioid misuse risk play a role?
- What counterfactual policies can decrease both opioid misuse and death?
- This paper highlights:
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  - How the perception of opioid misuse risk amplifies the policies' effects.

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- Develop a dynamic of work and opioid misuse with perception bias:
  - To quantify the heterogeneous substitution toward illegal opioids by policies.
  - To measure the magnitude of the perception bias on opioid misuse.
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In the past 12 months:

- Prescription user: was prescribed opioids and used them as directed
- Opioid misuser: used opioids not as directed by a doctor in any way
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- Opioid Use Disorder (OUD): suffered from using opioids including dependence, adverse symptoms, doing routine tasks at home, work, etc. (DSM-IV)

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- Laws on prescribing opioids:
  - Set a maximum number of days, amount, etc.
  - Are aggregated to represent stringency for all prescriptions on adults.
  - 2015: 5 states, 2019: 39 states including D.C.
- Must-Access Prescription Drug Monitoring Program (MA-PDMP)
  - Requires doctors to check patient's prescription history.
  - Decreases prescription on the extensive margin. (Sacks et al. [2021])
  - 2015: 16 states, 2019: 34 states including D.C.
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Data

- Opioid use and labor
  - **NSDUH**: Restricted National Survey of Drug Use and Health, 2015-2019
  - **NVSS**: Restricted National Vital Statistics System, 2000-2019
  - **MEPS**: Medical Expenditure Panel Survey, 2015-2019
  - **SIPP**: Survey of Income and Program Participation, 2014 and 2019 panel.
- Policies on opioid prescribing and opioid prices
  - **PDAPS**: Prescription Drug Abuse Policy System
  - **StreetRx**: 2013-2019

Data Patterns

Fact 1: Opioid misuse is associated with displacement from labor and perception of opioid misuse risk

	No Work & Illegal	
Displacement from Labor		
Laid off	0.888	(0.455)
Unable to Work	1.218	(0.458)
Other		
Opi: No Great Risk	0.557	(0.124)
Bad Health	0.135	(0.124)
College	0.010	(0.161)
Observations	18602	

Controls include age categories and a constant. Standard errors in parentheses. The sample is adults over 22 who **are** displaced from labor and **NOT** prescribed opioids.

Table: Multinomial Logit Regression of Work and Opioid Misuse, Public NSDUH, 2015-2019.

Back logit

Data Patterns

Fact 1: Opioid misuse is associated with displacement from labor and perception of opioid misuse risk

	No Work & Rx		No Work & Both	
Displacement from Labor				
Laid off	1.506	(0.321)	0.899	(1.066)
Unable to Work	1.171	(0.296)	0.056	(1.074)
Other				
Opi: No Great Risk	0.450	(0.152)	1.052	(0.173)
Bad Health	0.403	(0.144)	0.309	(0.182)
College	-0.117	(0.199)	-0.071	(0.250)
Observations	12591			

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Fact 1: Opioid misuse is associated with displacement from labor and perception of opioid misuse risk

	No Work & Illegal		Work & No Illegal		Work & Illegal	
Opi: No Great Risk	0.344	(0.158)	-0.218	(0.035)	0.206	(0.070)
Bad Health	0.774	(0.157)	-0.751	(0.042)	-0.209	(0.089)
College	-0.854	(0.189)	0.605	(0.029)	-0.009	(0.068)
Observations	103980					

Controls include age categories and a constant. Standard errors in parentheses. The sample is adults over 22 who are **NOT** displaced from labor and **NOT** prescribed opioids.

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Fact 1: Opioid misuse is associated with displacement from labor and perception of opioid misuse risk

	No Work & Rx		No Work & Both		Work & No Misuse		Work & Rx		Work & Both	
Opi: No Great Risk	0.462	(0.178)	0.572	(0.218)	-0.219	(0.059)	0.209	(0.096)	0.665	(0.111)
Bad Health	0.337	(0.158)	0.312	(0.228)	-0.786	(0.050)	-0.401	(0.102)	-0.228	(0.141)
College	-0.113	(0.189)	-0.467	(0.336)	0.609	(0.047)	0.741	(0.081)	0.047	(0.115)
Observations	45219									

Controls include age categories and a constant. Standard errors in parentheses. The sample is adults over 22 who are **NOT** displaced from labor and **are** prescribed opioids.

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Back mlogit

Data Patterns

Fact 2: The perception of opioid misuse risk is associated with displacement from labor

	Opi: Not Great Risk	
Perception of Drug Use Risk		
Coc: No Great Risk	2.399	(0.034)
LSD: No Great Risk	2.229	(0.037)
Mrj: No Great Risk	0.065	(0.058)
Opioid Use		
First-time Misuser	0.110	(0.128)
Never Used	0.248	(0.033)
Prescribed	-0.191	(0.027)
Displacement from Labor		
Laid Off	0.242	(0.048)
Unable	0.318	(0.066)
Retired	0.129	(0.078)
Observations	180392	

Controls include bad health, education, age categories, and a constant. Standard errors in parentheses. The sample is adults over 22.

Table: Logit Regression of Perceiving misusing Opioids as Not a Great Risk, Public NSDUH, 2015-2019.

Data Patterns

Fact 3: Opioid Misuse and Overdose Deaths are Prevalent amongst Working Age

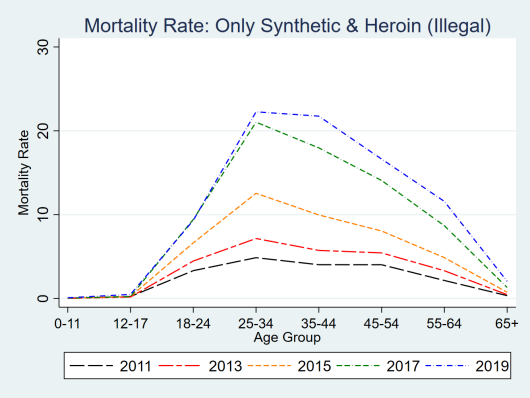

Figure: Overdose deaths: Illegal Only

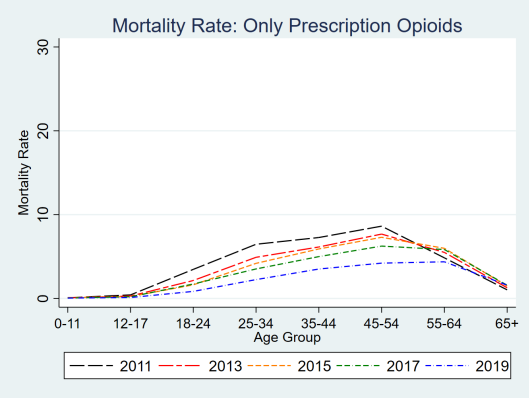

Figure: Overdose deaths: Prescription Only

Back Misuse

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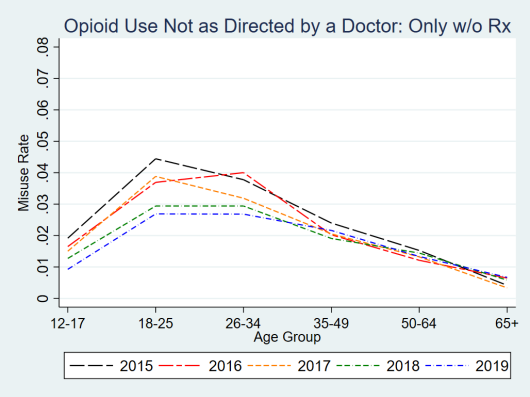

Figure: Opioid misuse: Illegal Only

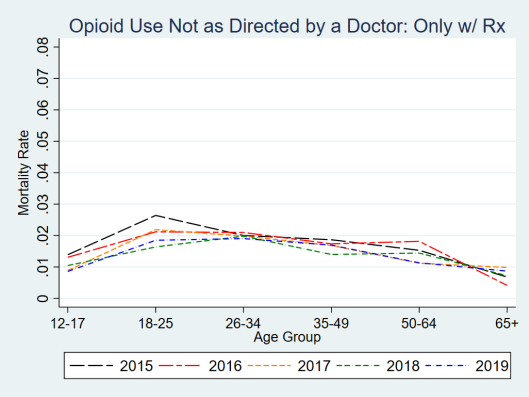

Figure: Opioid misuse: Prescription Only

Back OD

Data Patterns

- State-level restrictions on prescribing opioids are:
  - Associated with decreases in opioid dispense rate Reg
  - Associated with higher opioid overdose death rate by illegal opioids Reg
  - Not associated with prices for illegally traded opioids Reg
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Dynamic Model of Work and Opioid Use

Overview

- A representative individual lives in state g with education e.
- Life begins at age $a = 23$ and faces a risk of dying every period.
- Once he reaches 65, he stops aging and solves an infinite-horizon problem until he dies.

Figure: State Realization and Decision in Each Period

$$\Omega_g = (g, r_{g,t}, p_t^{rx}, p_{g,t}^{il})$$

$$\Omega_a = (e, a, exp, s, cw, bh, rx, \pi, \varepsilon)$$

State Space

Dynamic Model of Work and Opioid Use

Overview

- In each period, the individual observes his state's policies on prescribing opioids $r_{g,t}$ and opioid prices $p_t^{rx}, p_{g,t}^{il}$.
- He observes his work experience exp and opioid use disorder s.

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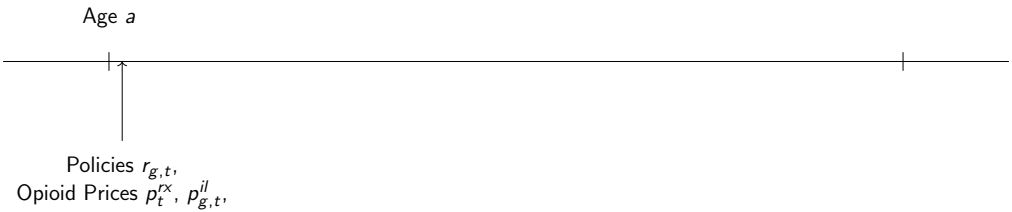

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Dynamic Model of Work and Opioid Use

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- Displacement from labor *cw* and bad health *bh* are realized.
- *cw*: 1=unemployed, 2=unable to work due to health conditions, 3=retired.
- He is also prescribed opioids *rx* conditional on his state and policies.

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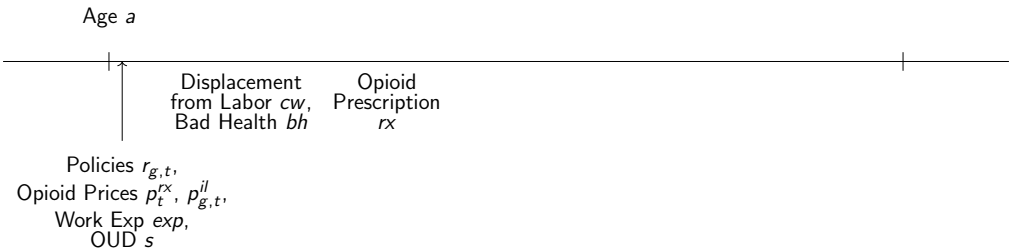

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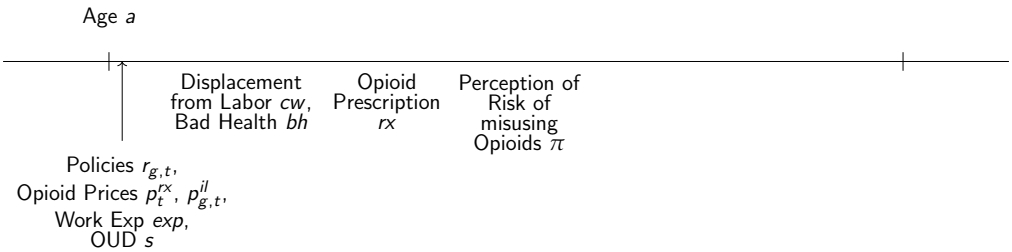
Dynamic Model of Work and Opioid Use

Overview

- The perception of opioid misuse risk π is realized conditional on his state.
- $\pi \in \{H, L\}$, H: high opioid misuse risk, L: Low risk opioid misuse risk



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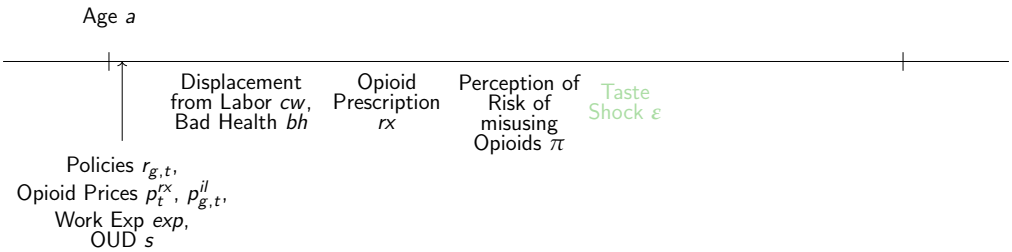

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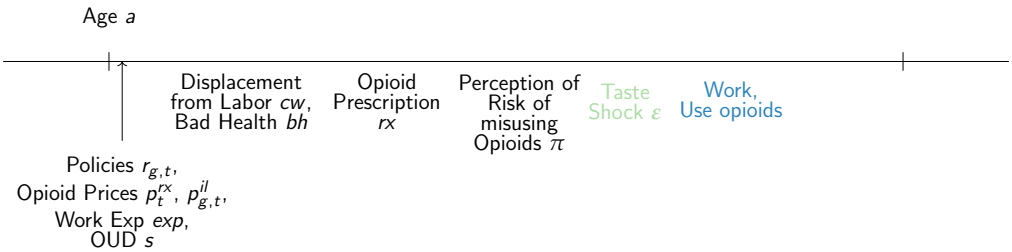

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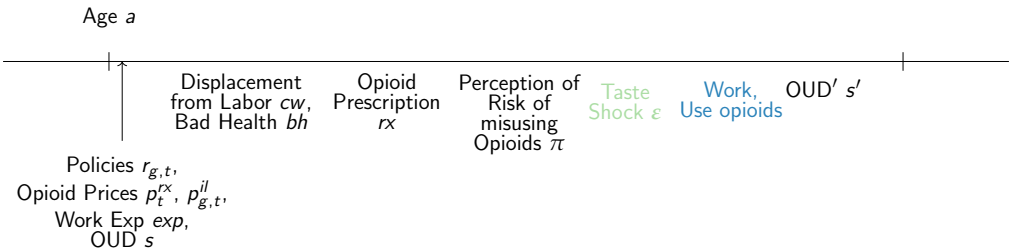

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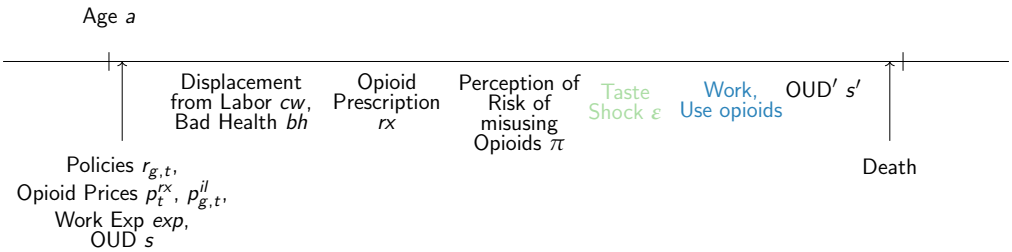

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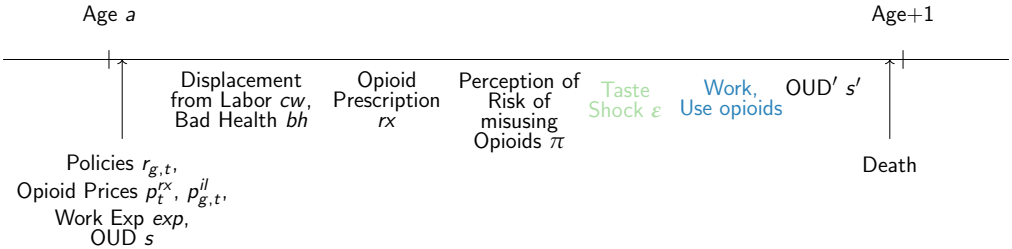
State Space

Dynamic Model of Work and Opioid Use

Overview

- If he survives, he ages a year.
- From 60, people can be retired from work.
  - He is retired next period if he does not work or is displaced from labor.
  - Once retired, he expects he will never return to work.



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Dynamic Model of Work and Opioid Use

Choice Set

- The individual can work and misuse opioids each period:

$$d_w = \mathbf{1}\{\text{work}\}$$
$$d_o^{rx} = \mathbf{1}\{\text{misuse prescribed opioids}\}$$
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- The choice set is affected by i) displacement from labor *cw* and ii) being prescribed opioids *rx*.
  - If he is displaced from labor, then he cannot work in this period.
  - He can misuse prescribed opioids only if he is prescribed opioids.
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Formal Table

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Dynamic Model of Work and Opioid Use

Per-period Utility

- Denote x_a the set of individual state variables except s and π.

$$u_j(x_a, s, \pi, \Omega_g, \varepsilon; \boldsymbol{\theta}^y, \boldsymbol{\theta}^w, \boldsymbol{\theta}^o) =$$
$$\underbrace{\bar{y}(x_a; \boldsymbol{\theta}^y)}_{\text{Baseline Income}} + \underbrace{y(x_a, s, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^w) d_w}_{\text{Income from Working}} + \underbrace{u_o(x_a, \Omega_g, s, d_w, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^o)}_{\text{Utility from misusing opioids}} + \varepsilon_j.$$

- Baseline income approximates lifetime earnings.
- Income from work reflects productivity loss from OUD and opioid misuse.
- Utility from misusing opioids captures:
  - Compensating for the disutility of being displaced from labor
  - The disutility of not misusing opioids if one has Opioid Use Disorder
  - The complementarity of misusing opioids and not working

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Dynamic Model of Work and Opioid Use

Transition Probabilities

- Suffering from OUD s'
- Death
- Displacement from work & bad health (cw', bh')
- Being prescribed opioids rx'
- Perception of the risk of misusing opioids π'

$$f_s(s, rx, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^s) = \begin{cases} 0 & \text{if } rx = d_o^{rx} = d_o^{il} = 0 \\ \frac{\exp(x_s \boldsymbol{\theta}^s)}{1+\exp(x_s \boldsymbol{\theta}^s)} & \text{o.w.} \end{cases}$$

where $x_s = [s, rx, d_o^{rx}, d_o^{il}, d_o^{rx} d_o^{il}]$

Dynamic Model of Work and Opioid Use

Transition Probabilities

- Suffering from OUD s'
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$$f_d(s', a, e, d_{rx}, d_{il}; \theta^d) = \sum_{i \in D} f(dth_i | s', a, e, g, d_{rx}, d_{il}; \boldsymbol{\theta}_i^d)$$

$$D = \{\text{OCD}, \text{OD:Rx}, \text{OD:Illegal}, \text{OD:Rx \& Illegal}\}$$

Dynamic Model of Work and Opioid Use

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$$f_{ch}((cw', bh') = i | x_{ch}; \boldsymbol{\theta}^{ch})$$

$$= \frac{\exp(x_{ch}\theta_i^{ch} + \alpha_g^{ch})}{1 + \sum_{i=1}^{7} \exp(x_{ch}\theta_i^{ch} + \alpha_g^{ch})}, i \neq (0,0)$$

$$x_{ch} = [\mathbf{1}_{cw=1}, \mathbf{1}_{cw=2}, \mathbf{1}_{cw=3}, bh, rx, a, e, exp, d_w, s']$$

$\alpha_g^{ch}$is a state-FE.

Dynamic Model of Work and Opioid Use

Transition Probabilities

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$$f_{rx}(x_{rx}, \alpha_g, r_g; \boldsymbol{\theta}^{rx}) = \frac{\exp(x_{rx}\theta^{rx} + \gamma r_g + \alpha_g)}{1 + \exp(x_{rx}\theta^{rx} + \gamma r_g + \alpha_g^{rx})}$$

$x_{rx} = [\mathbf{1}_{cw'=1}, \mathbf{1}_{cw'=2}, \mathbf{1}_{cw'=3}, bh', rx, a, e, s']$

α_g^{rx} is a state-FE.

Dynamic Model of Work and Opioid Use

Transition Probabilities

- Suffering from OUD s'
- Death
- Displacement from work & bad health (cw', bh')
- Being prescribed opioids rx'
- Perception of the risk of misusing opioids π'

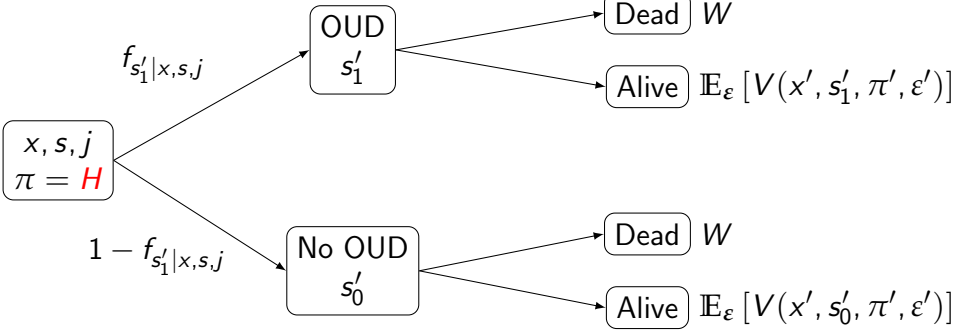
$$f_\pi(x_\pi; \boldsymbol{\theta}^\pi) = \frac{\exp(x_\pi \theta^\pi)}{1 + \exp(x_\pi \theta^\pi)}$$

$$x_\pi = [\mathbf{1}_{cw'=1}, \mathbf{1}_{cw'=2}, \mathbf{1}_{cw'=3}, bh', rx', s']$$

Dynamic Model of Work and Opioid Use

Perception of the risk of misusing opioids

- The individual stochastically perceives the risk of misusing opioids, π.
  - High risk ($\pi = H$): expects the actual probability of getting OUD
  - Low risk ($\pi = L$): expects a lower probability of getting OUD by δ



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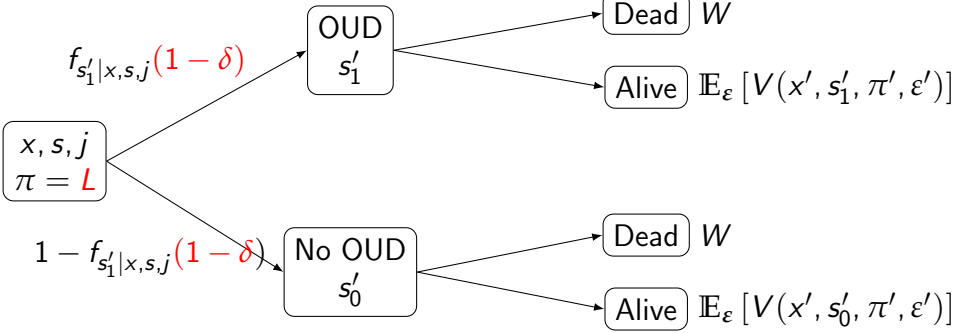

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Dynamic Model of Work and Opioid Use

Value Function Representation

- The value function of a living individual $V(x, s, \pi, \boldsymbol{\varepsilon})$ is

$$V(x_a, s, \pi, \boldsymbol{\varepsilon}) = \max_{j \in \mathcal{J}(cw,rx)} u_j(x_a, s; \boldsymbol{\theta}^y, \boldsymbol{\theta}^o) + \varepsilon_j +$$

$$\beta \left[\begin{array}{r} (1-\pi\delta) f_{s_1'|x_a,s,j} \left(f_{d|x_a,s_1',j} W + (1 - f_{d|x_a,s_1',j}) \sum_{x_a',\pi'} \mathbb{E}_\varepsilon \left[V(x_a', s_1', \pi', \boldsymbol{\varepsilon}') \right] f_{x_a'|x_a,s_1',d_j} \right) \\ +[1-(1-\pi\delta) f_{s_1'|x_a,s,j})] \left(f_{d|x_a,s_0',j} W + (1 - f_{d|x_a,s_0',j}) \sum_{x_a',\pi'} \mathbb{E}_\varepsilon \left[V(x_a', s_0', \pi', \boldsymbol{\varepsilon}') \right] f_{x_a'|x_a,s_0',d_j} \right) \end{array} \right]$$

- W: the (dis)utility upon death
- δ: the magnitude of discounting the probability of getting opioid use disorder
- The model is solved by contraction mapping at 65 and backward induction.

Alternative

Identification

The parameters to identify are:

- Initial type probability $\bar{q}(s|x_a)$
- Per-period utility
  - Income process θ^y, σ_y
  - Utility from misusing opioids θ^o
- Transition probabilities
  - Opioid Use Disorder θ^s
  - Death θ^d
  - Displacement from work and bad health θ^{ch}, α_g^{ch}
  - Prescription to opioids θ^{rx}, γ, α_g^{rx}
  - Perception of the risk of misusing opioids θ^{π}
- (Dis)utility of death W
- Perception bias δ
- Discount factor $\beta = 0.96$
- Distribution of ε: T1EV

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Identification

The parameters to identify are:

- Initial type probability $\bar{q}(s|x_a)$
- Per-period utility
  - Income process θ^y, σ_y
  - Utility from misusing opioids θ^o
- Transition probabilities
  - Opioid Use Disorder θ^s
  - Death θ^d
  - Displacement from work and bad health θ^{ch}, α_g^{ch}
  - Prescription to opioids θ^{rx}, γ, α_g^{rx}
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Identification

First stage

If $\bar{q}(s|x_a)$ is known, the parameters and CCPs below are identifiable:

- Perception of the risk of misusing opioids $\boldsymbol{\theta}^{\pi}$
- Unrestricted CCPs $P_j(x_a, s, \pi)$
- The income process $\boldsymbol{\theta}^{y}$, σ_y
- Transition to Opioid Use Disorder $\boldsymbol{\theta}^{s}$

Then, the transition parameters below are identifiable

- Death $\boldsymbol{\theta}^{d}$
- Displacement from labor and bad health $\boldsymbol{\theta}^{ch}$, α_g^{ch}
- Prescription to opioids $\boldsymbol{\theta}^{rx}$, γ, α_g^{rx}

Challenge

Identification

Challenge

- s is not observed; use four proxies to identify $\bar{q}(s|x_a)$
  - $\pi_1^* \in \{0,1\}$: "great risk" of using marijuana
  - $\pi_2^* \in \{0,1\}$: "great risk" of using cocaine
  - $\pi_3^* \in \{0,1\}$: "great risk" of using LSD
  - $ev \in \{0,1\}$: "Ever misused opioids in the past"
- NSDUH '15-'19 does not report s' if $d_o^{rx} = d_o^{il} = 0$ even when $rx = 1$:
  - Use NSDUH '21 for $f_s(s'|rx = 1, d_o^{rx} = 0, d_o^{il} = 0)$
- Assumption on the proxy variables:
  - They have logistic probabilities conditional on (x_a, s).
  - They are independent conditional on (x_a, s).
- Then, $\bar{q}(s|x_a)$, f_{ev}, f_{π_1}, f_{π_2}, and f_{π_3} are identified. (Hwang [2021])

Transition

Identification

- NSDUH is an RCS: need auxiliary panel data for $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, α_g^{ch}, α_g^{rx}
- NSDUH + NVSS, MEPS, and SIPP
  - NVSS: Death $\boldsymbol{\theta}^d$
  - SIPP: Constraint to work & bad health $\boldsymbol{\theta}^{ch}$
  - MEPS: Opioid Prescription $\boldsymbol{\theta}^{rx}$
  - NSDUH 2015-2019: Year-by-year distribution

Details Limited Feedback

Identification

Hotz-Miller Inversion and Finite Dependence

(Dis)utility of death W, Utility from misusing opioids θ^o, Perception bias δ

- Hotz-Miller inversion + finite dependence (Arcidiacono and Miller [2019])
- Challenge: δ is unknown and affects finite dependence paths.

Proposition

If there exists a 1-period finite dependence path (FD) for a subset of state transitions where the perception stayed "great risk" for two periods ($\pi = H, \pi' = H$), then a finite dependence path exists as a known function of δ under the rank condition in Arcidiacono and Miller [2019].

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Identification

- $\boldsymbol{\theta}^o, W, \delta$ identified from a set of conditional value function (CVF) differences with finite dependence, given all other parameters for transition probabilities and CCPs ▸ HM-Inversion-FD
- The variations for identification are:
  - $\boldsymbol{\theta}^o$: CVF differences at (x, s) between opioid misuse
  - W: Differences-in-CVF differences between opioid misuse across ages
  - δ: Differences-in-CVF differences between opioid misuse across different π's

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Estimation

A modified 'two-step' CCP estimation (Arcidiacono and Miller 2011, 2019):

Stage 1: Estimate CCPs and the transition probabilities.

1. EM algorithm: θ^u, $P_j(x, s, \pi)$, θ^π, θ^y, θ^s Details
2. Indirect inference: θ^d, θ^{ch}, θ^{rx} given the EM algorithm estimates Details

Stage 2: Estimate θ^o, W, δ.

1. Choose a set of conditional value function differences. Details
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Estimation

- Current version uses $a = 25, 45, 65$ with 3 states g. $\approx$30,000 moments.
- Initial guess on $\delta = 0.15$.
- Transition probabilities are sensitive to the targeted marginal transition probabilities.
- The second stage is computationally heavy.

Estimation

- The first stage estimates with smaller covariates:
  - Show reasonable signs and magnitudes on income, but not for being prescribed opioids.
  - The estimated s.d. of the measurement error on income is large.
- Calibration exercises with plausible transitions in the second stage show:
  - The coefficients for the disutility of death and opioid misuse are noisy, but Opioid Use Disorder is sensible.
  - Increase in the prescription rates does not necessarily increase future opioid misuse, but this is sensitive to the rate of getting Opioid Use Disorder θ_s.

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Conclusion

- This paper evaluates policy interventions on opioid prescribing.
  - Develops a dynamic model to quantify the substitution toward illegal opioids.
  - Focuses on opioid misuse and labor for policy's heterogeneous effect.
  - Highlights the role of the perception of opioid misuse risk.
- Model validation: compare with reduced-form results in the literature.
- Predict responses to counterfactual policies.
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Conclusion

"When you add hopelessness, unemployment, and lack of education on top of all that, it's kind of a recipe for disaster."

- Jan Rader, Huntington Fire Department Chief, *Heroin(e)*

Identification

- Suppose we have a FD path by only using tomorrow's state where $\pi = 0$:

$$\kappa(x_a'', s'', \pi_0 | x_a, s, \pi_0, i; \pi_0') = \kappa(x_a'', s'', \pi_0 | x_a, s, \pi_0, j; \pi_0')$$

- Then, the FD path by only using tomorrow's state where $\pi' = 1$ is:

$$\omega(x_a', s', \pi_1' | x_a, s, \pi_0, i) = g\begin{pmatrix} \delta, \kappa(x_a'', s'', \pi_0'' | x_a, s, \pi_0, i; \pi_0'), \\ f(s_1 | x, s, \pi_0)), f(x_a', s', \pi_1' | x_a, s, \pi_0, i) \end{pmatrix}$$

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Back

Utility Function in Detail

Each term in the utility function has the following functional form:

$$
\begin{aligned}
\bar{y}(x_a;\boldsymbol{\theta}^y) &= \theta_1^y + \theta_2^y e + \theta_3^y \frac{a}{10} + \theta_4^y (\frac{a}{10})^2 + \theta_5^y \mathbf{1}\{a=65\} + \sum_{k=1}^{3} \theta_{k+5}^y \mathbf{1}\{cw=k\} \\
&+ \theta_9^y \frac{a}{10} e + \theta_{10}^y (\frac{a}{10})^2 e + \theta_{11}^y \mathbf{1}\{a=65\} e + \sum_{k=1}^{3} \theta_{k+11}^y \mathbf{1}\{cw=k\} e \\
y(x_a, s, d_o^{rx}, d_o^{il};\boldsymbol{\theta}^w) &= \theta_1^w + \theta_2^w e + \theta_3^w \frac{a}{10} + \theta_4^w (\frac{a}{10})^2 + \theta_5^w \mathbf{1}\{a=65\} \\
&+ \theta_6^w \frac{a}{10} e + \theta_7^w (\frac{a}{10})^2 e + \theta_8^w \mathbf{1}\{a=65\} e \\
&+ \theta_9^w bh + \theta_{10}^w s + \theta_{11}^w bh\, rx + \theta_{12}^w bh\, d_o^{rx} + \theta_{13}^w d_o^{il} + \theta_{14}^w d_o^{rx}\, d_o^{il} \\
u_o(x_a, \Omega_g, s, d_w, d_o^{rx}, d_o^{il}) &= \sum_{k=1}^{2} \mathbf{1}\{cw=k\} (\theta_{3k-2}^o d_o^{il} + \theta_{3k-1}^o d_o^{rx} + \theta_{3k}^o\, d_o^{rx}\, d_o^{il}) \\
&+ bh(\theta_7^o d_o^{il} + \theta_8^o d_o^{rx} + \theta_9^o d_o^{rx}\, d_o^{il}) \\
&+ s(\theta_{10}^o (1-d_o^{il}) + \theta_{11}^o (1-d_o^{rx}) + \theta_{12}^o (1-d_o^{rx})\,(1-d_o^{il})) \\
&+ (1-d_w)(\theta_{13}^o d_o^{il} + \theta_{14}^o d_o^{rx} + \theta_{15}^o d_o^{rx}\, d_o^{il}) \\
&+ \theta_{16} rx\, p_t^{rx} + \theta_{17} d_o^{il} p_{g,t}^{il}
\end{aligned}
$$

Back

Opioid Rx, Misuse, OUD, and Overdose

	12-17	18-25	26-34	35-49	50-64	65+
Nonuser	81.56	70.76	66.77	65.47	62.51	64.07
Rx User	15.29	22.39	26.62	29.95	34.17	34.55
Misuser:						
- no OUD	2.71	5.80	5.24	3.65	2.65	1.20
- OUD	0.44	1.05	1.37	0.93	0.67	0.18
Total	100.00	100.00	100.00	100.00	100.00	100.00
Death						
Opioid Overdose	0.0005	0.0125	0.0262	0.0241	0.0180	0.0034
Other Causes of Death	0.0256	0.0733	0.1005	0.2054	0.7244	4.0101
Total	0.0261	0.0858	0.1267	0.2294	0.7424	4.0135

Table: Percentages of Nonuser, Rx User, Misuser, OUD, and Overdose deaths, NSDUH and NVSS: 2015-2019

Back By Year

Opioid Rx, Misuse, OUD, and Overdose

	2015	2016	2017	2018	2019
Nonuser	61.88	64.36	65.07	66.74	68.73
Rx User	33.29	31.11	30.70	29.37	27.44
Misuser:					
- no OUD	3.90	3.68	3.41	3.13	3.18
- OUD	0.93	0.85	0.81	0.77	0.64
Total	100.00	100.00	100.00	100.00	100.00
Death					
Opioid Overdose	0.0131	0.0166	0.0186	0.0181	0.0191
Other Causes of Death	1.0506	1.0492	1.0653	1.0674	1.0643
Total	1.0637	1.0658	1.0839	1.0855	1.0835

Table: Percents of Nonuser, Rx User, Misuser, and OUD. NSDUH & NVSS: 2015-2019

Back By Age

Opioid Death Rates across Age

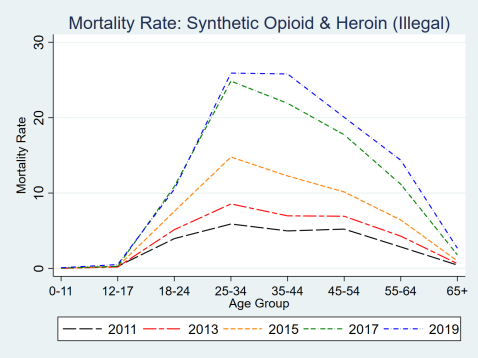


Figure: Mortality rates by illegal opioids across age groups, NVSS, 2011-2019

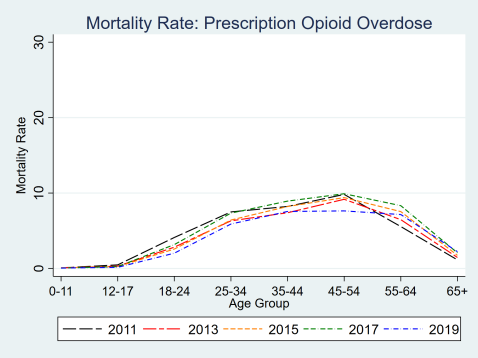


Figure: Mortality rates by prescription opioids across age groups, NVSS, 2011-2019

Back

Opioid Use and Constraints to Work

- Constraint to work is associated with opioid use
  - "laid off": higher misuse of opioids
  - "unable to work": higher Rx

	No Distruption	Unable to Work	Laid Off	All
Nonuser	67.00	35.87	64.96	65.38
Rx user	29.12	57.41	26.52	30.36
Misuser:				
- no OUD	3.26	4.55	5.93	3.46
- OUD	0.62	2.17	2.59	0.80
Total	100.00	100.00	100.00	100.00

Table: Percentages of Nonuser, Rx User, Misuser, and OUD by constraints to work, NSDUH PUF: 2015-2019

Row Back

Opioid Use and Constraints to Work

	No Distruption	Unable to Work	Laid Off	All
Nonuser	92.21	2.66	5.13	100.00
Rx User	86.32	9.17	4.51	100.00
Misuse:				
- no OUD	84.76	6.38	8.86	100.00
- OUD	70.11	13.17	16.72	100.00
Total	89.99	4.85	5.16	100.00

Table: Percentages of Nonuser, Rx User, Misuser, and OUD by constraints to work, NSDUH PUF: 2015-2019

Col Back

State-level Restrictions and Rx Opioids

- I use aggregate data to proxy for opioid Rx rate ($y_{i,t}$):
  - Rx/100: Opioid Rx per 100 population in state g in year t
  - MME[1]: The amount of opioids dispensed in state g in year-**quarter** t
- Two-way fixed effects (TWFE) regression with 2000-2019 data:

$$\log(y_{g,t}) = \beta_0 r_{g,t} + \sum_{k=1}^{3} \beta_j f\{cw = k\}_{g,t} + \beta_4 f\{bh\}_{g,t} + \alpha_g + \delta_t + \varepsilon_{g,t}$$

- The coefficient β are negative for both regressions.

Back

State-level Restrictions and Illegal Opioids

Prices for illegally traded opioids: 2014-2019

- Are street prices correlated with the state-level restrictions on prescription opioids?
- Average of the reported price per MME in state g and year t during 2013-2019

$$\bar{p}_{g,t} = \beta_0 r_{g,t} + \sum_{k=1}^{3} \beta_j f\{cw = k\}_{g,t} + \beta_4 f\{bh\}_{g,t} + \alpha_g + \delta_t + \varepsilon_{g,t}$$

- The coefficient β is statistically insignificant.

Back

State-level Restrictions and Mortality Rates

Mortality rates by types of opioid

- Mortality Rate = Number of Deaths / Population * 100,000
- Two-way fixed effect regression:

$$\text{Mortality Rate}_{g,t} = \beta_0 r_{g,t} + \sum_{k=1}^{3} \beta_j f\{cw = k\}_{g,t} + \beta_4 f\{bh\}_{g,t} + \alpha_g + \delta_t + \varepsilon_{g,t}$$

- The coefficients β are positive for synthetic opioids, negative for heroin, and (insignificant) positive for Rx opioids.

Back

Choice Set

- An individual can choose to work or misuse opioids each period:

$$d_w = \mathbf{1}\{\text{work}\}$$
$$d_o^{rx} = \mathbf{1}\{\text{misuse Rx opioids}\}$$
$$d_o^{il} = \mathbf{1}\{\text{use illegal opioids}\}$$

- An action is indexed by $j = 1 + d_o^{rx} + 2d_o^{il} + 4d_w$ and $d_j = \mathbf{1}\{\text{Chooses } j\}$.
- The choice set $\mathcal{J}(rx, k)$ is defined by i) being prescribed opioids rx and ii) displacement from labor market $k = \mathbf{1}\{cw \neq 0\}$:

$$\mathcal{J}(0,0) = \{1,3,5,7\}$$
$$\mathcal{J}(1,0) = \{1,2,4,5,6,8\}$$
$$\mathcal{J}(0,1) = \{1,3\}$$
$$\mathcal{J}(1,1) = \{1,2,4\}$$

Back Table State Space

Choice Set

- An individual can choose to work or misuse opioids each period:

$$d_w = \mathbf{1}\{\text{work}\}$$
$$d_o^{rx} = \mathbf{1}\{\text{misuse Rx opioids}\}$$
$$d_o^{il} = \mathbf{1}\{\text{use illegal opioids}\}$$

- Denote $\tilde{d}_j$ as a negation of action d_j.

(cw, rx)	d_w	$d_o^{rx}, \tilde{d}_o^{il}$	$\tilde{d}_o^{rx}, d_o^{il}$	d_o^{rx}, d_o^{il}
(0,0)	✓	✗	✓	✗
(0,1)	✓	✓	✗	✓
(1,0)	✗	✗	✓	✗
(1,1)	✗	✓	✗	✓

Table: Choice Set

Back Formal State Space

Dynamic Model of Work and Opioid Use

State Variables

- Age $a = 23, \ldots, 65$, state g
- State-level restrictions $r_g \in \{0, 1\}$, Rx opioid price p^{rx}, illegal opioid price p_g^{il}
- Work experience $exp \in \{0, 1\}$, education $e \in \{0, 1\}$
- Bad health $bh \in \{0, 1\}$
- Displacement from work: $cw \in \{0, 1, 2, 3\}$
  - 1: laid off, 2: ill/disabled, 3: retired.
  - $k := \mathbf{1}\{cw \neq 0\}$
- Prescribed opioids $rx \in \{0, 1\}$, OUD $s \in \{0, 1\}$
- Perception of the risk of misusing opioids $\pi \in \{0, 1\}$
- Idiosyncratic shocks to preference $\boldsymbol{\varepsilon} := (\varepsilon_1, \ldots, \varepsilon_{J(rx,k)})$

Choice Set

Derivation: CCP

$$P(d_j|x,\pi,\pi^*,ev) = \sum_s q(s|x,\pi,\pi^*,ev)P(d_j|x,\pi,s)$$

$$= \sum_s \frac{P(\pi|x,s,\pi^*,ev;\boldsymbol{\theta}^\pi)q(s|x,\pi^*,ev)}{P(\pi|x,\pi^*,ev)}P(d_j|x,\pi,s) \quad \because \text{Bayes' Rule}$$

$$= \sum_s \frac{P(\pi|x,s;\boldsymbol{\theta}^\pi)q(s|x,\pi^*,ev)}{P(\pi|x,\pi^*,ev)}P(d_j|x,\pi,s) \quad \because \text{Assumption}$$

Back

Derivation: Transition Prob for OUD

$$
\begin{aligned}
&P_s(s'|x, \pi, \pi^*, ev, d_j) \\
&= \sum_s q(s|x, \pi, \pi^*, ev, d_j) f_s(s'|s, rx, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^s) \\
&= \sum_s \frac{P(d_j|s, x, \pi, \pi^*, ev)}{P(d_j|x, \pi, \pi^*, ev)} q(s|x, \pi, \pi^*, ev) f_s(s'|s, rx, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^s) \\
&= \sum_s \frac{P(d_j|s, x, \pi)}{P(d_j|x, \pi, \pi^*, ev)} \frac{P(\pi|x, s; \boldsymbol{\theta}^\pi) q(s|x, \pi^*, ev)}{P(\pi|x, \pi^*, ev)} f_s(s'|s, rx, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^s)
\end{aligned}
$$

Back

Identification: Unrestricted CCPs

$$P(d_j|x,\pi,\pi^*,ev) = \sum_s \frac{P(\pi|x,s;\boldsymbol{\theta}^\pi)q(s|x,\pi^*,ev)}{P(\pi|x,\pi^*,ev)} P(d_j|x,\pi,s)$$

- Let $iv = (\pi^*, ev)$ and suppose $iv \in \{0,1\}$. For a given (d_j, x, π, iv) and $q(s|x,iv)$:

$$\underbrace{\begin{bmatrix} q(s_1|x,iv_0) & 1-q(s_1|x,iv_0) \\ q(s_1|x,iv_1) & 1-q(s_1|x,iv_1) \end{bmatrix}}_{:=Q} \underbrace{\begin{bmatrix} P(d_j|x,s_1) \\ P(d_j|x,s_0) \end{bmatrix}}_{:=p_s} = \underbrace{\begin{bmatrix} P(d_j|x,iv_0) \\ P(d_j|x,iv_1) \end{bmatrix}}_{:=p_x}$$

- If $q(s_1|x,iv_0) \neq q(s_1|x,iv_1)$, then $det(Q) \neq 0$. Thus, $p_s = Q^{-1}p_x$.
- With more variation in iv, $q(s|x,iv)$ is over-identified.

Back

Probability of Dying

- The probability of dying increases wrt (d_o^{rx}, d_o^{il}):

$$f_d(s', a, d_{rx}, d_{il}; \theta^d) = \sum_{i \in D} f(dth_i | s', a, g, d_{rx}, d_{il}; \boldsymbol{\theta}_i^d)$$

where $D = \{ocd, rx, il, bth\}$

$$\begin{aligned}
f(dth_{ocd} | s', a, g, d_{rx}, d_{il}; \theta^{ocd}) &= \frac{\exp(x'_{ocd}\theta^d)}{1 + \exp(x'_{ocd}\theta^d)} \\
f(dth_{rx} | s', a, g, d_{rx}, d_{il}; \theta^{rx}) &= \theta^d_{rx} d_{rx} \\
f(dth_{il} | s', a, g, d_{rx}, d_{il}; \theta^{il}) &= \theta^d_{il} d_{il} \\
f(dth_{bth} | s', a, g, d_{rx}, d_{il}; \theta^{bth}) &= \theta^d_{bth} d_{rx} d_{il} \\
f(dth | s') &= \theta^d_s s'
\end{aligned}$$

Transition

Id: Transition Probabilities

NVSS, NSDUH, SIPP, MEPS

$$P_d^{ocd}(a,g) = \sum_{s'} f_d(a, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}_d) P(s', d_o^{rx} = 0, d_o^{il} = 0 | a, g)$$

$$P_d^{OD:rx} = \sum_{s', d_o^{il}} f_d(a, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}_d) P(s', d_o^{rx} = 1, d_o^{il})$$

$$P_d^{OD:il} = \sum_{s', d_o^{rx}} f_d(a, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il} = 1)$$

$$P_d^{OD:bth} = \sum_{s'} f_d(a, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}_d) P(s', d_o^{rx} = 1, d_o^{il} = 1)$$

Back

Id: Transition Probabilities

NVSS, NSDUH, SIPP, MEPS

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi}, rx, rx', s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \boldsymbol{\theta}_d) \\ f(cw', bh'|cw, bh, rx, d_j, s') \\ f(rx'|cw', bh', rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi}, s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \boldsymbol{\theta}_d) \\ f(cw', bh'|cw, bh, rx, d_j, s') \\ f(rx'|cw', bh', rx, d_j, s') \end{bmatrix}$$

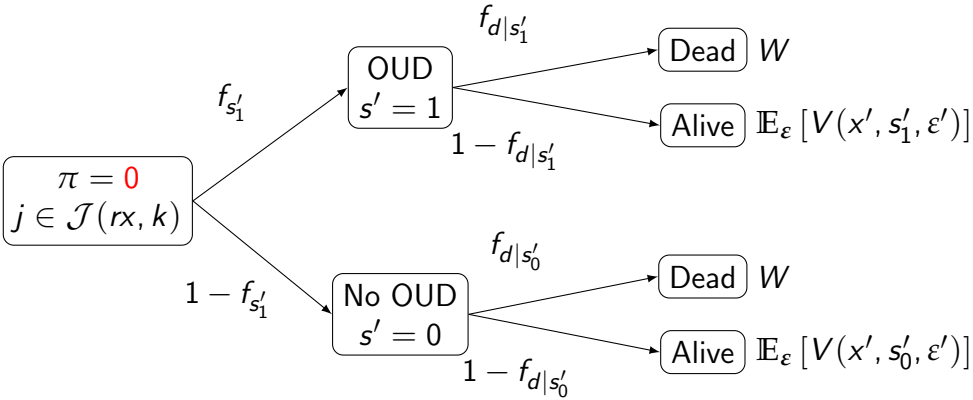
$$P(cw', rx', bh') = \sum_{cw, rx, bh, d_j, s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \boldsymbol{\theta}_d) \\ f(cw', bh'|cw, bh, rx, d_j, s') \\ f(rx'|cw', bh', rx, d_j, s') \end{bmatrix}$$

Back

Dynamic Model of Work and Opioid Use

Perception of the risk of misusing opioids

- In each period, the individual perceives the risk of misusing opioids, π.
  - High risk ($\pi = 0$): expects the actual probability of getting OUD
  - Low risk ($\pi = 1$): expects a lower probability of getting OUD



Simple

Dynamic Model of Work and Opioid Use

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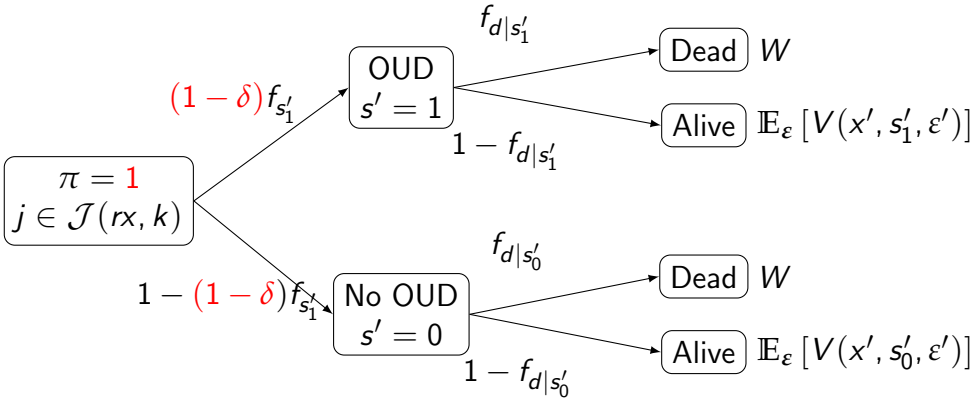

Simple

CVF Differences

- Hotz-Miller inversion implies:

$$\log \frac{P(d_j|x,s)}{P(d_1|x,s)} = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$

$$+ \beta \begin{bmatrix} (1-\pi\delta) f_{s_1',j} \left(f_{d|s_1',j} W + \sum_{x'} \bar{V}(x', s_1') f_{x'|x,s_1',j}(\boldsymbol{\theta}^{ch}, \boldsymbol{\theta}^{rx}) \right) \\ +[1-(1-\pi\delta) f_{s_1',j})] \left(f_{d|s_0',j} W + \sum_{x'} \bar{V}(x', s_0') f_{x'|x,s_0',j}(\boldsymbol{\theta}^{ch}, \boldsymbol{\theta}^{rx}) \right) \end{bmatrix}$$

$$- \beta \begin{bmatrix} (1-\pi\delta) f_{s_1',1} \left(f_{d|s_1',1} W + \sum_{x'} \bar{V}(x', s_1') f_{x'|x,s_1',1}(\boldsymbol{\theta}^{ch}, \boldsymbol{\theta}^{rx}) \right) \\ +[1-(1-\pi\delta) f_{s_1',1})] \left(f_{d|s_0',1} W + \sum_{x'} \bar{V}(x', s_0') f_{x'|x,s_0',1}(\boldsymbol{\theta}^{ch}, \boldsymbol{\theta}^{rx}) \right) \end{bmatrix}$$

Back

Derivation

- Rearranging the RHS:

$$\log \frac{P(d_j|x,s)}{P(d_1|x,s)} = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o)) \qquad \text{Today's Utility}$$

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix} \qquad \text{Death}$$

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix} \qquad \text{Tomorrow's Life with OUD}$$

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix} \qquad \text{Tomorrow's Life without OUD}$$

Value Function: Alternative

$$
\begin{aligned}
V(x_a, s, \pi, \varepsilon) = & \max_{j \in \mathcal{J}(cw,rx)} u_j(x_a, s; \boldsymbol{\theta}^y, \boldsymbol{\theta}^o) + \varepsilon_j \\
& + \beta \underbrace{Wf_{d|x_a,s_1',j}(1-\pi\delta) f_{s_1'|x_a,s,j}}_{\text{Death with Opioid Use Disorder}} \\
& + \beta \sum_{x_a',\pi'} \underbrace{\bar{V}(x_a', s_1', \pi') f_{x'|x_a,s_1',j}(1 - f_{d|x_a,s_1',j})(1-\pi\delta) f_{s_1'|x_a,s,j}}_{\text{Tomorrow's Life with Opioid Use Disorder}} \\
& + \beta \underbrace{Wf_{d|x_a,s_0',j}(1-(1-\pi\delta) f_{s_1'|x_a,s,j})}_{\text{Death without Opioid Use Disorder}} \\
& + \beta \sum_{x_a',\pi'} \underbrace{\bar{V}(x_a', s_0', \pi') f_{x_a'|x_a,s_0',j}(1 - f_{d|x_a,s_0',j}) \left(1-(1-\pi\delta) f_{s_1'|x_a,s,j}\right)}_{\text{Tomorrow's Life without OUD}}
\end{aligned}
$$

Back

Derivation: CVF with future CVF weights

$$v_j(x,s,\pi) = u_j(x,s,\theta^o)$$ Flow utility in $t+1$

$$+\beta\left[W(1-\pi\delta)f_{s_1'|s,j}f_{d'|x,s_1',j} + W(1-(1-\pi\delta)f_{s_1'|s,j})f_{d'|x,s_0',j}\right]$$ Death in $t+1$

$$+\beta\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} u_k(x',s_1',\pi')+ \\ \psi_k(\mathbf{p}(x',s_1',\pi'))\end{bmatrix}\omega_{k,s_1'}f_{x',\pi'|x,s_1',j}(1-\pi\delta)f_{s_1'|x,s,j}\tilde{f}_{d'|x,s',j}$$ Flow utility w/ OUD in $t+1$

$$+\beta\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} u_k(x',s_0',\pi')+ \\ \psi_k(\mathbf{p}(x',s_0',\pi'))\end{bmatrix}\omega_{k,s_0'}f_{x',\pi'|x,s_0',j}(1-(1-\pi\delta)f_{s_1'|x,s,j})\tilde{f}_{d'|x,s',j}$$ Flow utility w/o OUD in $t+1$

$$+\beta^2\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} W(1-\pi'\delta)f_{s_1''|s_1',k}f_{d''|x',s_1'',k}+ \\ W(1-(1-\pi'\delta)f_{s_1''|s_1',k})f_{d''|x',s_0'',k}\end{bmatrix}\omega_{k,s_1'}f_{x',\pi'|x,s_1',j}(1-\pi\delta)f_{s_1'|x,s,j}\tilde{f}_{d'|x,s',j}$$ Death w/ OUD in $t+1$

$$+\beta^2\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} W(1-\pi'\delta)f_{s_1''|s_0',k}f_{d''|x',s_1'',k}+ \\ W(1-(1-\pi'\delta)f_{s_1''|s_0',k})f_{d''|x',s_0'',k}\end{bmatrix}\omega_{k,s_0'}f_{x',\pi'|x,s_1',j}(1-(1-\pi\delta)f_{s_1'|x,s,j})\tilde{f}_{d'|x,s',j}$$ Death w/o OUD in $t+2$

$$+\beta^2\sum_{x'',s''}\sum_{x'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} V(x'',s_1'',\pi'')f_{x''|x',s_1',k}f_{d''|s_1''}(1-\pi')f_{s_1''|s_1'}+ \\ V(x'',s_0'',\pi'')f_{x''|x',s_0',k}f_{d''|s_0''}(1-(1-\pi'))f_{s_1''|s_1'}\end{bmatrix}\omega_{k,s_0'}f_{x|s_1'}(1-\pi\delta)f_{s_1'|s,j}\tilde{f}_{d|s',j}$$ Survival w/ OUD in $t+2$

$$+\beta^2\sum_{x'',s''}\sum_{x'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} V(x'',s_1'',\pi'')f_{x''|x',s_1',k}f_{d''|s_1''}(1-\pi')f_{s_1''|s_0'}+ \\ V(x'',s_0'',\pi'')f_{x''|x',s_0',k}f_{d''|s_0''}(1-(1-\pi'))f_{s_1''|s_0'}\end{bmatrix}\omega_{k,s_1'}f_{x|s_0'}(1-(1-\pi\delta)f_{s_1'|s,j})\tilde{f}_{d|s',j}$$ Survival w/o OUD $t+2$

where $\omega_{k,s_1'} = \omega_k(x',s_1',\pi',x,s,\pi,j)$ and $\omega_{k,s_0'} = \omega_k(x',s_0',\pi',x,s,\pi,j)$

Back

CVF Differences with Finite Dependence

If a pair of choices $(1, j)$ at (x, s, π) exhibit a 1-finite dependence, then:

$$
\begin{aligned}
&\psi_1(\mathbf{p}(x,s,\pi)) - \psi_j(\mathbf{p}(x,s,\pi)) = v_j(x,s,\pi) - v_1(x,s,\pi) \\
&= u_j(x,s;\boldsymbol{\theta}^y,\boldsymbol{\theta}^o) - u_1(x,s;\boldsymbol{\theta}^y,\boldsymbol{\theta}^o) && \Delta \text{ Flow utility in } t \\
&+ \beta\left[W(1-\pi\delta) f_{s_1'|s,j} f_{d'|x,s_1',j} + W(1-(1-\pi\delta) f_{s_1'|s,j}) f_{d'|x,s_0',j}\right] && \text{Death in } t+1 \text{ given } j \\
&- \beta\left[W(1-\pi\delta) f_{s_1'|s,1} f_{d'|x,s_1',j} + W(1-(1-\pi\delta) f_{s_1'|s,1}) f_{d'|x,s_0',1}\right] && \text{Death in } t+1 \text{ given } 1 \\
&+ \beta \sum_{x',\pi'} \sum_{k\in\mathcal{J}(x')} \begin{bmatrix} u_k(x',s_1';\boldsymbol{\theta}^y,\boldsymbol{\theta}^o) + \\ \psi_k(\mathbf{p}(x',s_1',\pi')) \end{bmatrix} \omega_{k,s_1',j} f_{x',\pi'|x,s_1',j}(1-\pi\delta) f_{s_1'|x,s,j} \tilde{f}_{d'|x,s',j} && \text{Flow utility w/ OUD in } t+1 \text{ given } j \\
&- \beta \sum_{x',\pi'} \sum_{k\in\mathcal{J}(x')} \begin{bmatrix} u_k(x',s_1';\boldsymbol{\theta}^y,\boldsymbol{\theta}^o) + \\ \psi_k(\mathbf{p}(x',s_1',\pi')) \end{bmatrix} \omega_{k,s_1',1} f_{x',\pi'|x,s_1',j}(1-\pi\delta) f_{s_1'|x,s,1} \tilde{f}_{d'|x,s',1} && \text{Flow utility w/ OUD in } t+1 \text{ given } 1 \\
&+ \beta \sum_{x',\pi'} \sum_{k\in\mathcal{J}(x')} \begin{bmatrix} u_k(x',s_0';\boldsymbol{\theta}^y,\boldsymbol{\theta}^o) + \\ \psi_k(\mathbf{p}(x',s_0',\pi')) \end{bmatrix} \omega_{k,s_0',j} f_{x',\pi'|x,s_0',j}(1-(1-\pi\delta) f_{s_1'|x,s,j}) \tilde{f}_{d'|x,s',j} && \text{Flow utility w/o OUD in } t+1 \text{ given } j \\
&- \beta \sum_{x',\pi'} \sum_{k\in\mathcal{J}(x')} \begin{bmatrix} u_k(x',s_0';\boldsymbol{\theta}^y,\boldsymbol{\theta}^o) + \\ \psi_k(\mathbf{p}(x',s_0',\pi')) \end{bmatrix} \omega_{k,s_0',1} f_{x',\pi'|x,s_0',1}(1-(1-\pi\delta) f_{s_1'|x,s,1}) \tilde{f}_{d'|x,s',1} && \text{Flow utility w/ OUD in } t+1 \text{ given } 1
\end{aligned}
$$

where $\omega_{k,s',j} = \omega_k(x', s', \pi', x, s, \pi, j; \delta)$

Back

Estimation: Stage 1-1

- The likelihood of observing $(x_n, \pi_n, d_{j,n}, y_{k,n}, s'_n, ev_n, \pi^*_n)$ is

$$l(x_n, \pi_n, d_{j,n}, y_n, s'_n; \boldsymbol{\theta}) =$$

$$\sum_{s=0,1} q(s|x_n, ev, \pi^*; \boldsymbol{\theta}^u) \prod_{j,\pi,y_k,s'} \begin{bmatrix} f_\pi(\pi|x_\pi, s; \boldsymbol{\theta}^\pi) \times \\ P(d_{j,n}|x_n, \pi_n, s) \times \\ F_y(y_{k,n}|d_{j,n}, x_n, s; \boldsymbol{\theta}^y, \sigma_y) \times \\ f_s(s'|d_{j,n}, rx_n, s; \boldsymbol{\theta}^s) \end{bmatrix}^{\mathbf{1}\{\pi_n, d_{jn}, y_{kn}, s'_n\}}$$

where $F_y(y_k|d_{j,n}, x_n, s; \boldsymbol{\theta}^y) = \Phi\left(\frac{y_{k+1,n} - \bar{y}(x_y, d_j; \boldsymbol{\theta}^y)}{\sigma_y}\right) - \Phi\left(\frac{y_{k,n} - \bar{y}(x_y, d_j; \boldsymbol{\theta}^y)}{\sigma_y}\right)$

Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
- NVSS, NSDUH, SIPP, MEPS

$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\theta_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
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$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
- NVSS, NSDUH, SIPP, MEPS

$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
- NVSS, NSDUH, SIPP, MEPS

$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
- NVSS, NSDUH, SIPP, MEPS

$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$ Today's Utility

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix}$$ Death

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix}$$ Tomorrow's Life with OUD

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix}$$ Tomorrow's Life without OUD

Identification Derivation Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$ Today's Utility

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix}$$ Death

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix}$$ Tomorrow's Life with OUD

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix}$$ Tomorrow's Life without OUD

Identification Derivation Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$ Today's Utility

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix}$$ Death

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix}$$ Tomorrow's Life with OUD

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix}$$ Tomorrow's Life without OUD

Identification Derivation Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$ Today's Utility

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix}$$ Death

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix}$$ Tomorrow's Life with OUD

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix}$$ Tomorrow's Life without OUD

Identification Derivation Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$
\begin{aligned}
&\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s) \\
&= \left(\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)\right) - \left(\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o)\right) && \text{Today's Utility} \\
&+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j}\left(f_{d|s_1',j} - f_{d|s_0',j}\right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1}\left(f_{d|s_1',1} - f_{d|s_0',1}\right) \end{bmatrix} && \text{Death} \\
&+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix} && \text{Tomorrow's Life with OUD} \\
&+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j}\right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1}\right) \end{bmatrix} && \text{Tomorrow's Life without OUD}
\end{aligned}
$$

Identification Derivation Back

Estimation: Stage 2

- For a given $\boldsymbol{\omega}(x', s', x, s)$ where $\sum_{k \in \mathcal{J}(rx', k')} \omega_k(x', s', x, s) = 1$ and $|\omega_k(x', s', x, s)| < \infty$,

$$\bar{V}(x', s_0') = \sum_{k \in \mathcal{J}(rx', k')} \left(v_k(x', s_0') - \log P(d_k | x', s_0')\right) \omega_k(x', s_0', x, s)$$

$$\bar{V}(x', s_1') = \sum_{k \in \mathcal{J}(rx', k')} \left(v_k(x', s_1') - \log P(d_k | x', s_1')\right) \omega_k(x', s_1', x, s)$$

- Iterate $v_k(x', s')$ once more wrt utilities and future value functions
- Find the weights $\omega_k(x'', s_0'', x, s)$ that cancels out the two-period-ahead ex-ante value functions $\bar{V}(x'', s'')$
- Then, we have a moment for identifying $\boldsymbol{\theta}^o$.

Back Back

Estimation: Stage 3

- Suppose we have such weights for each $(x, s, j, 1)$.
- Then, estimating $\boldsymbol{\theta}^o, W, \delta$ becomes a GMM problem:

$$\begin{bmatrix} \log P(d_2|x_1, s_0) - \log P(d_1|x_1, s_0) \\ \vdots \\ \log P(d_J|x_1, s_0) - \log P(d_1|x_1, s_0) \\ \log P(d_2|x_1, s_1) - \log P(d_1|x_1, s_1) \\ \vdots \\ \log P(d_J|x_1, s_1) - \log P(d_1|x_1, s_1) \\ \vdots \\ \log P(d_2|x_M, s_1) - \log P(d_1|x_M, s_1) \\ \vdots \\ \log P(d_J|x_M, s_1) - \log P(d_1|x_M, s_1) \end{bmatrix} = \begin{bmatrix} v_2(x_1, s_0) - v_1(x_1, s_0) \\ \vdots \\ v_J(x_1, s_0) - v_1(x_1, s_0) \\ v_2(x_1, s_1) - v_1(x_1, s_1) \\ \vdots \\ v_J(x_1, s_1) - v_1(x_1, s_1) \\ \vdots \\ v_2(x_M, s_1) - v_1(x_M, s_1) \\ \vdots \\ v_J(x_M, s_1) - v_1(x_M, s_1) \end{bmatrix}$$

Back

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